

## Problem

An array is a simple data structure used to store a collection of data in a contiguous block of memory. Each element in the collection is accessed using an index, and the elements are easy to find because they're stored sequentially in memory.

## Submissions

Because the collection of elements in an array is stored as a big block of data, we typically use arrays when we know exactly how many pieces of data we're going to have. For example, you might use an array to store a list of student ID numbers, or the names of state capitals. To create an array of integers named **myArray** that can hold four integer values, you would write the following code:

```
int[] myArray = new int[4];
```

## Leaderboard

This sets aside a block of memory that's capable of storing **4** integers. Each integer storage cell is assigned a unique index ranging from **0** to one less than the size of the array, and each cell initially contains a **0**. In the case of **myArray**, we can store integers at indices **0**, **1**, **2**, and **3**. Let's say we wanted the last cell to store the number **12**; to do this, we write:

```
myArray[3] = 12;
```

## Discussions

Similarly, we can print the contents of the last cell with the following code:

```
System.out.println(myArray[3]);
```

## Editorial

The code above prints the value stored at index **3** of **myArray**, which is **12** (the value we previously stored there). It's important to note that while Java initializes each cell of an array of integers with a **0**, not all languages do this.

## Task

The code in your editor does the following:

1. Reads an integer from stdin and saves it to a variable, **n**, denoting some number of integers.
2. Reads **n** integers corresponding to **a<sub>0</sub>, a<sub>1</sub>, ..., a<sub>n-1</sub>** from stdin and saves each integer **a<sub>i</sub>** to a variable, **val**.
3. Attempts to print each element of an array of integers named **a**.

Write the following code in the unlocked portion of your editor:

1. Create an array, **a**, capable of holding **n** integers.
2. Modify the code in the loop so that it saves each sequential value to its corresponding location in the array. For example, the first value must be stored in **a<sub>0</sub>**, the second value must be stored in **a<sub>1</sub>**, and so on.

Good luck!

## Input Format

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Language

Java 7



```
5 1 import java.util.*; ...
9
10 scan.close(); ...
```

Line: 9 Col: 1

Upload Code as File

Run Code

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☐ Test against custom input
